

Zachary Land

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Education

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| Imperial College London | Sep 2024 – Jul 2028 |
| <ul style="list-style-type: none">• Course: Joint Mathematics and Computer Science• Awarded Grade: 82.7% average; expected 1st class• Modules: Analysis, Multivariable Calculus, Numerical Linear Algebra, Programming, Statistical Modelling | |
| Hereford Sixth Form College & Whitecross Hereford High School | Sep 2017 – Jul 2024 |
| <ul style="list-style-type: none">• Courses: Computer Science, Further Mathematics, Mathematics, Physics• Awarded Grades: A*A*A*A*; 10 grade 9 GCSEs | |

Experience

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| SC26 International Student Cluster Competition | Jul 2026 – Nov 2026 |
| <ul style="list-style-type: none">• Selected to represent the UK for the international student cluster competition in November.• Optimising matrix multiplication, NAMD, and LINPACK for faster execution ahead of the competition. | |
| Google Research Engineer Intern | Jun 2026 – Sep 2026 |
| <ul style="list-style-type: none">• Working on classifier probes for LLM content safety.• Working on automating the creation of malicious prompts for Gemini. | |
| Aon Risk Modeller Intern | Jun 2025 – Aug 2025 |
| <ul style="list-style-type: none">• Implemented a statistical model to forecast catastrophic losses and analyse market risk.• Self-taught VBA to innovate Excel macros, reducing data science pipeline execution time by 10 minutes. | |

Projects

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| vLLM KV Cache Quantization | May 2026 – Jun 2026 |
| <ul style="list-style-type: none">• Expanded LLM context length 4x by quantizing vLLM KV cache from FP16 to blockwise NVFP4.• Preserved vLLM paging efficiency and used QAT distillation to quantize the transformer to FP4. | |
| ICHack Winner | Feb 2026 |
| <ul style="list-style-type: none">• Won ICHack, Europe's largest student-run hackathon, with a mathematical model to predict oil spill trajectories.• Formulated ODEs and monte-carlo simulations on a graph lattice to calculate routes with minimal ecological risk. | |
| Metaphor Alignment Research Paper | Feb 2026 – Apr 2026 |
| <ul style="list-style-type: none">• Authored a CCN-accepted extended abstract on detecting conceptual metaphor alignment in embedding spaces.• Developed a graph-based metric that identified metaphor alignments missed by CKA. | |
| UCL AI Research Hackathon Winner | Nov 2025 – Jul 2026 |
| <ul style="list-style-type: none">• Presented a machine learning model with representational alignment to identify discrepancies in LLM output.• Researching an extension of the project in collaboration with, and funded by, Holistic AI. | |
| Probabilistic Computing Research | Oct 2025 – Jan 2026 |
| <ul style="list-style-type: none">• Built a probabilistic numerical linear algebra package and an Ising model to expedite OPM-MEG brain scanning.• Awarded best undergraduate research project in Oxford's probabilistic computing research competition. | |
| IBM-Z Datathon Winner | Oct 2025 |
| <ul style="list-style-type: none">• Competed against 6500+ participants to develop a real-time riot prediction model.• Applied NLP with PCA for sentiment analysis and a GNN to forecast unrest through influence networks. | |

Skills

Programming Languages: Python, C++, C, SQL, VBA, C#, Java, Kotlin, Haskell
Languages: English (native), Spanish (A2), Japanese & Mandarin (elementary)

Achievements & Extracurricular

Gold in the British Physics Computational Challenge; Gold in the British Physics Olympiad at the age of 16
Top 30 in the UK for Long Jump at the age of 14; V6-Level Climber